

GATE DA (Data Science & AI) Practice Question Bank - Set 10

4 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers four topics: Covariance, Correlation & the Central Limit Theorem, the k-NN Algorithm, SQL Aggregates/GROUP BY/HAVING, and Simple & Compound Interest.

SAME HONESTY POLICY AS BEFORE: For Topic 1, the covariance question is reconstructed from a documented GATE DA 2024 result ($E[X]=1/4$, $E[Y]=3/4$, $E[XY]=1/4$, $\text{Cov}=1/16=0.063$) into a fully self-consistent joint-PMF table, since the exact original table wasn't recoverable -- it is labelled accordingly rather than claimed as verbatim. CLT itself is a confirmed, heavily-tested GATE DA topic. For Topic 2, GATE DA 2024 is confirmed to have tested k-NN with a 2-mark figure-based question, but the exact figure/data wasn't recoverable, so self-contained worked examples are used instead, labelled [GATE-style]. For Topic 3, the HAVING/GROUP BY true-false question is a genuine, well-documented GATE CS 2012 question (re-derived from first principles below), and the correlated-subquery aggregate question is adapted from GATE CSE 2021. Topic 4 (Simple & Compound Interest) is standard aptitude content with no claim to a specific verified PYQ.

Section B ("New Practice Questions") in every topic is freshly written practice, as always.

Contents

1. Topic 1: Covariance, Correlation & the Central Limit Theorem — Section A (Old/PYQ-style, 10 Qs) + Section B (New Practice, 10 Qs)
2. Topic 2: k-NN Algorithm — Section A (Old/PYQ-style, 10 Qs) + Section B (New Practice, 10 Qs)
3. Topic 3: SQL - Aggregates, GROUP BY, HAVING — Section A (Old/PYQ-style, 10 Qs) + Section B (New Practice, 10 Qs)
4. Topic 4: Simple & Compound Interest — Section A (Old/PYQ-style, 10 Qs) + Section B (New Practice, 10 Qs)

Topic 1: Covariance, Correlation & the Central Limit Theorem

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, reconstructed from a documented GATE DA 2024 covariance result]

Q1. [NAT · 2 marks]

X and Y are binary (0/1) random variables with joint PMF: $P(X=0,Y=0)=1/4$, $P(X=0,Y=1)=1/2$, $P(X=1,Y=0)=0$, $P(X=1,Y=1)=1/4$. Find $\text{Cov}(X,Y)$ (correct to 3 decimals).

Answer: 0.063

Solution: $E[X] = 0 \times (1/4 + 1/2) + 1 \times (0 + 1/4) = 1/4$. $E[Y] = 0 \times (1/4 + 0) + 1 \times (1/2 + 1/4) = 3/4$. $E[XY] = 1 \times 1 \times P(X=1,Y=1) = 1/4$. $\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = 1/4 - (1/4)(3/4) = 1/4 - 3/16 = 1/16 = 0.0625 \approx 0.063$.

[GATE-style, confirmed core GATE DA topic]

Q2. [MCQ · 1 mark]

Which theorem explains why the distribution of sample means tends toward a normal distribution as sample size increases, regardless of the population's original distribution (under mild conditions)?

- (A) Bayes' Theorem
- (B) Central Limit Theorem
- (C) Law of Total Probability
- (D) Markov's Inequality

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

If X and Y are independent random variables, then:

- (A) $\text{Cov}(X,Y) = 0$
- (B) $\text{Var}(X+Y) = \text{Var}(X) - \text{Var}(Y)$
- (C) $E[X|Y] = E[X] \times E[Y]$
- (D) $\text{Cov}(X,Y) = \text{Var}(X)$

Answer: (A)

[GATE-style]

Q4. [NAT · 1 mark]

According to the Central Limit Theorem, if a population has variance $\sigma^2=100$, what is the variance of the sample mean for a sample of size $n=25$?

Answer: 4

Solution: $\text{Var}(\text{sample mean}) = \sigma^2/n = 100/25 = 4$.

[GATE-style]

Q5. [MCQ · 1 mark]

As the sample size n increases, the standard error of the sample mean ($\sigma/\sqrt{n}$):

- (A) Increases
- (B) Decreases

- (C) Stays the same
- (D) Becomes negative

Answer: (B)

[GATE-style]

Q6. [NAT · 1 mark]

A population has mean $\mu=50$ and standard deviation $\sigma=20$. For a sample size of $n=400$, find the standard error of the sample mean.

Answer: 1

Solution: $20/\sqrt{400} = 20/20 = 1$.

[GATE-style]

Q7. [MCQ · 1 mark]

The Central Limit Theorem applies to the distribution of:

- (A) Individual data points from any population
- (B) The SAMPLE MEAN (or sum) of a sufficiently large number of i.i.d. random variables
- (C) Only normally distributed populations
- (D) Only discrete random variables

Answer: (B)

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following are TRUE about covariance and correlation?

- (A) $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$
- (B) Correlation $\rho = \text{Cov}(X,Y)/(\sigma_X\sigma_Y)$
- (C) Correlation is a scaled, unit-free version of covariance, bounded in $[-1,1]$
- (D) Covariance is always bounded between -1 and 1 , just like correlation

Answer: (A), (B), (C)

Solution: (D) is false -- unlike correlation, covariance has no fixed bounds; its magnitude depends on the scale/units of X and Y .

[GATE-style]

Q9. [MCQ · 1 mark]

If $\text{Var}(X)=9$ and $\text{Var}(Y)=16$, and X, Y are independent, find $\text{Var}(X+Y)$.

- (A) 25
- (B) 5
- (C) 7
- (D) 144

Answer: (A)

Solution: $\text{Var}(X+Y) = \text{Var}(X)+\text{Var}(Y) = 9+16 = 25$ (since independent).

[GATE-style]

Q10. [NAT · 1 mark]

Using the previous question's values (independent, $\text{Var}(X)=9$, $\text{Var}(Y)=16$), find the standard deviation of $X+Y$.

Answer: 5

Solution: $\sqrt{25} = 5$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

X,Y are binary with joint PMF: $P(0,0)=0.1$, $P(0,1)=0.3$, $P(1,0)=0.2$, $P(1,1)=0.4$. Find $E[X]$.

Answer: 0.6

Solution: $E[X] = 0(0.1+0.3) + 1(0.2+0.4) = 0.6$.

[New Practice]

Q12. [NAT · 1 mark]

Using the same PMF, find $E[Y]$.

Answer: 0.7

Solution: $E[Y] = 0(0.1+0.2) + 1(0.3+0.4) = 0.7$.

[New Practice]

Q13. [NAT · 1 mark]

Using the same PMF, find $E[XY]$.

Answer: 0.4

Solution: $E[XY] = 1 \times 1 \times P(1,1) = 0.4$.

[New Practice]

Q14. [NAT · 2 marks]

Using the results from the previous three questions, find $\text{Cov}(X,Y)$.

Answer: -0.02

Solution: $\text{Cov}(X,Y) = 0.4 - (0.6 \times 0.7) = 0.4 - 0.42 = -0.02$.

[New Practice]

Q15. [MCQ · 1 mark]

A sample of size $n=64$ is drawn from a population with $\sigma=16$. According to the CLT, the sampling distribution of the sample mean is approximately normal with standard deviation (standard error):

- (A) 2
- (B) 4
- (C) 8
- (D) 16

Answer: (A)

Solution: $16/\sqrt{64} = 16/8 = 2$.

[New Practice]

Q16. [MSQ · 2 marks]

Which of the following are generally associated with the Central Limit Theorem?

- (A) The approximation to normality generally improves as sample size n increases
- (B) It applies regardless of the shape of the original population distribution (under mild regularity conditions, e.g. finite variance)

- (C) It requires the original population to already be normally distributed
- (D) It justifies using z-based confidence intervals/hypothesis tests for sample means, even for non-normal populations, when n is reasonably large

Answer: (A), (B), (D)

Solution: (C) is false -- the entire point of the CLT is that it works for a WIDE variety of underlying population distributions, not just normal ones.

[New Practice]

Q17. [NAT · 2 marks]

A fair six-sided die has $\text{Var}(X)=35/12 \approx 2.9167$. For $n=35$ independent rolls, find the approximate variance of the sample mean (correct to 4 decimals).

Answer: 0.0833

Solution: $2.9167/35 \approx 0.0833$.

[New Practice]

Q18. [MCQ · 1 mark]

Two random variables X and Y satisfy $\text{Cov}(X,Y)=0$. Which of the following can we conclude with CERTAINTY?

- (A) X and Y are independent
- (B) X and Y have no linear relationship (zero linear correlation), though they might still be non-linearly dependent
- (C) X equals Y
- (D) $\text{Var}(X)=\text{Var}(Y)$

Answer: (B)

[New Practice]

Q19. [NAT · 1 mark]

X and Y have correlation $\rho=0.5$, with $\sigma_X=4$ and $\sigma_Y=9$. Find $\text{Cov}(X,Y)$.

Answer: 18

Solution: $\text{Cov}(X,Y) = \rho \times \sigma_X \times \sigma_Y = 0.5 \times 4 \times 9 = 18$.

[New Practice]

Q20. [MCQ · 1 mark]

If a random sample of size n is taken from ANY population with finite mean μ and finite variance σ^2 , the Central Limit Theorem states that, for sufficiently large n, the sampling distribution of the sample mean $\bar{X}$ is approximately:

- (A) $\text{Exponential}(1/\mu)$
- (B) $N(\mu, \sigma^2/n)$
- (C) $\text{Uniform}(0,1)$
- (D) Identical to the original population's distribution

Answer: (B)

Topic 2: k-NN Algorithm

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, confirmed: k-NN was tested with a figure-based question in GATE DA 2024]

Q1. [MCQ · 1 mark]

In the k-Nearest Neighbors (k-NN) algorithm, a new data point is classified by:

- (A) Fitting a line through all the training data
- (B) Finding the k closest training points (by some distance metric) and assigning the majority class among them
- (C) Computing the mean of all training labels, regardless of distance
- (D) Randomly selecting a label

Answer: (B)

[GATE-style]

Q2. [NAT · 2 marks]

Training points: A(1,1), B(2,2), C(6,6), D(7,7). For a test point at (3,3), using Euclidean distance, find the distance to point B (correct to 4 decimals).

Answer: 1.4142

Solution: $\sqrt{(3-2)^2 + (3-2)^2} = \sqrt{2} \approx 1.4142$.

[GATE-style]

Q3. [NAT · 2 marks]

Using the same points, find the Euclidean distance from the test point (3,3) to point C(6,6) (correct to 4 decimals).

Answer: 4.2426

Solution: $\sqrt{(6-3)^2 + (6-3)^2} = \sqrt{18} \approx 4.2426$.

[GATE-style]

Q4. [MCQ · 1 mark]

Given the training points A(1,1,class0), B(2,2,class0), C(6,6,class1), D(7,7,class1), and distances from test point (3,3) of approximately $A \approx 2.83$, $B \approx 1.41$, $C \approx 4.24$, $D \approx 5.66$, what class would k-NN assign for $k=1$?

- (A) class0
- (B) class1
- (C) Tie
- (D) Undefined

Answer: (A)

Solution: The nearest point is B (distance ≈ 1.41), whose class is class0.

[GATE-style]

Q5. [MCQ · 1 mark]

For $k=1$ (a special case of k-NN), the algorithm reduces to:

- (A) Always predicting the majority class in the entire training set
- (B) The simple "nearest neighbor" rule -- assign the class of the single closest training point
- (C) Linear regression

(D) k-means clustering

Answer: (B)

[GATE-style]

Q6. [MCQ · 1 mark]

A very LARGE value of k in k-NN (approaching the size of the training set) tends to result in:

- (A) A highly flexible, overfit decision boundary
- (B) An overly smooth decision boundary that may underfit, tending toward always predicting the overall majority class
- (C) Exactly the same result as $k=1$
- (D) An undefined classification

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

A very SMALL value of k (e.g. $k=1$) in k-NN tends to result in:

- (A) High bias, low variance (an overly smooth/simple boundary)
- (B) Low bias, high variance (a highly flexible/sensitive decision boundary, prone to overfitting noisy points)
- (C) Exactly the same result as a very large k
- (D) No classification at all

Answer: (B)

[GATE-style]

Q8. [MCQ · 1 mark]

Why is choosing an ODD value of k generally recommended for BINARY classification in k-NN?

- (A) It makes computation faster
- (B) It avoids ties in the majority vote among the k nearest neighbours
- (C) It guarantees 100% training accuracy
- (D) It is a strict mathematical requirement of the algorithm's definition

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about the k-NN algorithm?

- (A) It is a "lazy learner"/instance-based method, since it defers computation to prediction time rather than building an explicit model during training
- (B) It requires storing the entire (or a significant portion of the) training dataset for use at prediction time
- (C) Feature scaling (normalization) is generally important for k-NN, since it relies on distance metrics sensitive to feature scale
- (D) k-NN always has zero training time AND zero prediction time

Answer: (A), (B), (C)

Solution: (D) is false -- while k-NN has minimal explicit "training" computation, PREDICTION time can be substantial, since distances to many (or all) training points must be computed for each new query.

[GATE-style]

Q10. [MCQ · 1 mark]

Which distance metric is most commonly used by default in k-NN for continuous numerical features?

- (A) Hamming distance
- (B) Euclidean distance
- (C) Jaccard distance
- (D) Levenshtein (edit) distance

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Training points: A(0,0), B(1,1), C(5,5), D(6,6). For a test point at (2,2), find the Euclidean distance to point B (correct to 4 decimals).

Answer: 1.4142

Solution: $\sqrt{(2-1)^2 + (2-1)^2} = \sqrt{2} \approx 1.4142$.

[New Practice]

Q12. [NAT · 1 mark]

Using the same points, find the Euclidean distance from the test point (2,2) to point A(0,0) (correct to 4 decimals).

Answer: 2.8284

Solution: $\sqrt{4+4} = \sqrt{8} \approx 2.8284$.

[New Practice]

Q13. [MCQ · 2 marks]

Given the four training points (A, B, C, D as above, with A:class0, B:class0, C:class1, D:class1) and distances from the test point (2,2) of $A \approx 2.83$, $B \approx 1.41$, $C \approx 4.24$, $D \approx 5.66$, which three points are the $k=3$ nearest neighbours?

- (A) A, B, C
- (B) A, B, D
- (C) B, C, D
- (D) A, C, D

Answer: (A)

Solution: Sorted by distance: B(1.41), A(2.83), C(4.24), D(5.66) -- the nearest 3 are B, A, C.

[New Practice]

Q14. [NAT · 1 mark]

For the $k=3$ neighbours identified above (A: class0, B: class0, C: class1), what is the majority-vote predicted class (give as 0 or 1)?

Answer: 0

Solution: 2 votes for class0 (A, B) vs 1 vote for class1 (C) -- majority is class0.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are common techniques to choose a good value of k in k-NN?

- (A) Cross-validation, trying several values of k and selecting the one with the best validation performance
- (B) Always using $k=1$, regardless of the dataset
- (C) Using the square root of the number of training examples as a rough heuristic starting point
- (D) Using an elbow-method-style plot of validation error versus k

Answer: (A), (C), (D)

Solution: (B) is false -- $k=1$ is prone to overfitting and is not a generally recommended universal choice.

[New Practice]

Q16. [MCQ · 1 mark]

In k-NN REGRESSION (as opposed to classification), the predicted value for a new point is typically:

- (A) The majority class among the k nearest neighbours
- (B) The average (mean) of the target values of the k nearest neighbours
- (C) The maximum target value among all training points
- (D) Always zero

Answer: (B)

[New Practice]

Q17. [NAT · 1 mark]

For k-NN regression with $k=3$, the three nearest neighbours have target values 10, 14, and 18. Find the predicted value (simple average).

Answer: 14

Solution: $(10+14+18)/3 = 14$.

[New Practice]

Q18. [MCQ · 1 mark]

"Weighted k-NN" differs from standard (unweighted) k-NN in that it:

- (A) Ignores distance entirely
- (B) Gives closer neighbours a higher influence (weight) on the final prediction than farther neighbours, rather than treating all k neighbours equally
- (C) Only ever uses the single nearest neighbour
- (D) Cannot be used for regression

Answer: (B)

[New Practice]

Q19. [MCQ · 1 mark]

The "curse of dimensionality" is a known challenge for k-NN because:

- (A) It makes training impossibly slow
- (B) In very high-dimensional feature spaces, distances between points tend to become less meaningful/discriminative, as most points end up roughly equidistant from each other
- (C) It only affects decision trees, not k-NN
- (D) It has no real effect on k-NN's performance

Answer: (B)

[New Practice]

Q20. [NAT - 1 mark]

A dataset has 1000 training points. For a single test query, assuming a brute-force approach computes the distance to every training point once, how many distance computations are needed?

Answer: 1000

Topic 3: SQL – Aggregates, GROUP BY, HAVING

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CS 2012, classic]

Q1. [MCQ · 2 marks]

Which of the following statements about an SQL query are TRUE?

P: An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause.

Q: An SQL query can contain a HAVING clause only if it has a GROUP BY clause.

R: All attributes used in the GROUP BY clause must appear in the SELECT clause.

S: Not all attributes used in the GROUP BY clause need appear in the SELECT clause.

- (A) P and R
- (B) P and S
- (C) Q and R
- (D) Q and S

Answer: (B)

Solution: P is TRUE -- a HAVING clause with no GROUP BY simply treats the entire table as a single group. Q is FALSE (it directly contradicts P). R is FALSE -- a column can be used to GROUP BY without being included in the SELECT list. S is TRUE, correctly stating the opposite of R. So the true statements are P and S.

[GATE-style, adapted from a GATE CSE 2021 query pattern]

Q2. [MCQ · 2 marks]

Given relation emp(empId, name, gender, salary, deptId), the query:

```
SELECT deptId, COUNT(*) FROM emp WHERE gender='female' AND salary > (SELECT  
AVG(salary) FROM emp) GROUP BY deptId;
```

retrieves, for each department:

- (A) The total number of employees in that department
- (B) The number of FEMALE employees in that department who earn MORE than the OVERALL average salary (across the entire company)
- (C) The average salary of female employees
- (D) The number of male employees earning above average

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

Which of the following aggregate functions IGNORE NULL values by default in standard SQL (except COUNT(*), which counts all rows)?

- (A) None of them; all aggregate functions include NULLs
- (B) SUM, AVG, COUNT(column), MAX, and MIN all ignore NULL values in the specified column
- (C) Only COUNT(*) ignores NULLs
- (D) Only SUM ignores NULLs

Answer: (B)

[GATE-style]

Q4. [NAT · 1 mark]

Table Orders has Amount values for OrderID 1-5: 100, 200, NULL, 150, 50. Compute SUM(Amount).

Answer: 500

Solution: $100+200+150+50 = 500$ (the NULL is ignored).

[GATE-style]

Q5. [NAT · 1 mark]

Using the same data, compute COUNT(Amount) (counting non-NULL values only).

Answer: 4

Solution: 4 non-NULL values (the NULL is excluded).

[GATE-style]

Q6. [NAT · 1 mark]

Using the same data, compute AVG(Amount) (correct to 2 decimals).

Answer: 125

Solution: $500/4 = 125$.

[GATE-style]

Q7. [MCQ · 2 marks]

The correct LOGICAL order of SQL clause evaluation (not the written syntax order) is generally:

- (A) SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY
- (B) FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY
- (C) GROUP BY, FROM, WHERE, SELECT, HAVING, ORDER BY
- (D) HAVING, GROUP BY, WHERE, FROM, SELECT, ORDER BY

Answer: (B)

Solution: Although SQL is WRITTEN as SELECT...FROM...WHERE...GROUP BY...HAVING...ORDER BY, it is logically EVALUATED as FROM (and joins) first, then WHERE (row filtering), then GROUP BY (grouping), then HAVING (group filtering), then SELECT (column/expression computation), then ORDER BY.

[GATE-style]

Q8. [MCQ · 1 mark]

The key difference between WHERE and HAVING is that:

- (A) WHERE filters rows before grouping; HAVING filters GROUPS after aggregation (and can reference aggregate functions)
- (B) WHERE and HAVING are functionally identical in every situation
- (C) HAVING can only be used without GROUP BY
- (D) WHERE can reference aggregate functions directly, but HAVING cannot

Answer: (A)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about the GROUP BY clause?

- (A) It groups rows that share the same value(s) in one or more specified columns
- (B) It is commonly used together with aggregate functions like COUNT, SUM, AVG, MIN, MAX
- (C) Every column in the SELECT list that is not inside an aggregate function must generally appear in the GROUP BY clause (in standard SQL)
- (D) GROUP BY always sorts the final result set in ascending order automatically, making ORDER BY redundant

Answer: (A), (B), (C)

Solution: (D) is false -- GROUP BY groups rows but does not guarantee any particular final output order; ORDER BY should be used explicitly whenever a specific order is required.

[GATE-style]

Q10. [NAT · 2 marks]

Table Employees(dept, salary) has rows: ('Sales',100), ('Sales',200), ('IT',300), ('IT',150), ('IT',250). For the query "SELECT dept, COUNT(*) FROM Employees GROUP BY dept HAVING COUNT(*) > 2", how many department rows satisfy the HAVING condition?

Answer: 1

Solution: Sales count = 2 (not >2). IT count = 3 (>2, qualifies). Only 1 department (IT) satisfies.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Table Sales(product, amount) has rows: ('A',50), ('A',70), ('B',30), ('B',40), ('B',60), ('C',100). For "SELECT product, SUM(amount) FROM Sales GROUP BY product", find the sum for product 'B'.

Answer: 130

Solution: $30+40+60 = 130$.

[New Practice]

Q12. [NAT · 1 mark]

Using the same data, find the sum for product 'A'.

Answer: 120

Solution: $50+70 = 120$.

[New Practice]

Q13. [MCQ · 1 mark]

Using the same data (A=120, B=130, C=100), which product has the HIGHEST total sales amount?

- (A) A
- (B) B
- (C) C
- (D) Tie

Answer: (B)

[New Practice]

Q14. [NAT · 2 marks]

Using "SELECT product, COUNT(*) FROM Sales GROUP BY product HAVING COUNT(*) >= 2" on the same data (A has 2 rows, B has 3 rows, C has 1 row), how many product groups satisfy this condition?

Answer: 2

Solution: A qualifies ($2 \geq 2$), B qualifies ($3 \geq 2$), C does not ($1 < 2$). Total = 2.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following aggregate functions can be used inside a HAVING clause?

- (A) COUNT(*)
- (B) SUM(column)
- (C) AVG(column)
- (D) None; aggregate functions cannot be used in HAVING

Answer: (A), (B), (C)

[New Practice]

Q16. [MCQ · 2 marks]

Which query correctly finds departments having more than 5 employees, using proper SQL syntax?

- (A) SELECT dept FROM Employees WHERE COUNT(*) > 5 GROUP BY dept;
- (B) SELECT dept, COUNT(*) FROM Employees GROUP BY dept HAVING COUNT(*) > 5;
- (C) SELECT dept FROM Employees HAVING COUNT(*) > 5;
- (D) SELECT dept FROM Employees GROUP BY dept WHERE COUNT(*) > 5;

Answer: (B)

Solution: Aggregate functions like COUNT(*) cannot be used directly in a WHERE clause (since WHERE filters rows before aggregation); HAVING must be used instead, after GROUP BY.

[New Practice]

Q17. [NAT · 1 mark]

Table Students(class, marks) has rows: ('X',40), ('X',60), ('X',80), ('Y',90), ('Y',70). Find MAX(marks) for class 'X'.

Answer: 80

[New Practice]

Q18. [NAT · 1 mark]

Using the same data, find MIN(marks) for class 'Y'.

Answer: 70

[New Practice]

Q19. [MCQ · 1 mark]

The query "SELECT COUNT(DISTINCT dept) FROM Employees;" returns:

- (A) The total number of employees
- (B) The number of DISTINCT (unique) department values appearing in the table
- (C) The number of departments with more than one employee
- (D) Always zero

Answer: (B)

[New Practice]

Q20. [NAT · 1 mark]

Table Products(category, price) has 20 rows total. The query "SELECT category, AVG(price) FROM Products GROUP BY category" returns 5 distinct category groups. If each group has an equal number of rows, how many rows are in EACH group?

Answer: 4

Solution: $20/5 = 4$.

Topic 4: Simple & Compound Interest

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [NAT · 1 mark]

Find the simple interest on Rs. 5000 at 8% per annum for 3 years.

Answer: 1200

Solution: $SI = P \times R \times T / 100 = 5000 \times 8 \times 3 / 100 = 1200$.

[GATE-style]

Q2. [NAT · 1 mark]

Find the amount (principal + simple interest) for Rs. 8000 at 10% per annum for 2 years.

Answer: 9600

Solution: $SI = 8000 \times 10 \times 2 / 100 = 1600$. Amount = $8000 + 1600 = 9600$.

[GATE-style]

Q3. [MCQ · 2 marks]

A sum doubles itself in 8 years at simple interest. Find the rate of interest per annum.

- (A) 12.5%
- (B) 10%
- (C) 8%
- (D) 25%

Answer: (A)

Solution: Let $P=100$; doubling means $SI=100$. $SI = PRT/100 \Rightarrow 100 = 100 \times R \times 8 / 100 \Rightarrow 100 = 8R \Rightarrow R = 12.5\%$.

[GATE-style]

Q4. [NAT · 2 marks]

Find the compound interest on Rs. 10000 at 10% per annum for 2 years, compounded annually.

Answer: 2100

Solution: Amount = $10000 \times (1.1)^2 = 10000 \times 1.21 = 12100$. CI = $12100 - 10000 = 2100$.

[GATE-style]

Q5. [NAT · 1 mark]

Find the amount on Rs. 15000 at 20% per annum for 2 years, compounded annually.

Answer: 21600

Solution: $15000 \times (1.2)^2 = 15000 \times 1.44 = 21600$.

[GATE-style]

Q6. [MCQ · 1 mark]

For the same principal, rate, and time period (with time > 1 year), compound interest compared to simple interest is generally:

- (A) Always equal to simple interest
- (B) Greater than simple interest (since CI compounds on previously earned interest too)
- (C) Always less than simple interest

(D) Unrelated to simple interest

Answer: (B)

[GATE-style]

Q7. [NAT · 2 marks]

Find the difference between compound interest and simple interest on Rs. 5000 at 10% per annum for 2 years.

Answer: 50

Solution: $SI = 5000 \times 10 \times 2 / 100 = 1000$. $CI = 5000 \times (1.1)^2 - 5000 = 6050 - 5000 = 1050$. Difference = $1050 - 1000 = 50$.

[GATE-style]

Q8. [MCQ · 1 mark]

The formula for compound interest, compounded ANNUALLY, with principal P, rate R% per annum, for T years, gives the amount A as:

- (A) $A = P(1 + R/100)^T$
- (B) $A = P(1 + RT/100)$
- (C) $A = P \times R \times T / 100$
- (D) $A = P + RT$

Answer: (A)

[GATE-style]

Q9. [NAT · 2 marks]

Find the compound interest on Rs. 20000 at 5% per annum for 3 years, compounded annually (correct to 2 decimals).

Answer: 3152.5

Solution: $A = 20000 \times (1.05)^3 = 20000 \times 1.157625 = 23152.50$. $CI = 23152.50 - 20000 = 3152.50$.

[GATE-style]

Q10. [MCQ · 1 mark]

If interest is compounded HALF-YEARLY (semi-annually) at an annual rate of R%, for T years, the correct formula for the amount is:

- (A) $A = P(1 + R/100)^T$
- (B) $A = P(1 + (R/2)/100)^{2T}$
- (C) $A = P(1 + 2R/100)^{T/2}$
- (D) $A = P(1 + R/200)^T$

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Find the simple interest on Rs. 12000 at 6% per annum for 4 years.

Answer: 2880

Solution: $12000 \times 6 \times 4 / 100 = 2880$.

[New Practice]

Q12. [NAT · 1 mark]

Find the amount for Rs. 9000 at 12% per annum simple interest for 5 years.

Answer: 14400

Solution: $SI = 9000 \times 12 \times 5 / 100 = 5400$. Amount = $9000 + 5400 = 14400$.

[New Practice]

Q13. [MCQ · 2 marks]

A sum triples itself in 10 years at simple interest. Find the rate of interest per annum.

- (A) 20%
- (B) 15%
- (C) 30%
- (D) 10%

Answer: (A)

Solution: Tripling means $SI = 2P$. $2P = P \times R \times 10 / 100 \Rightarrow 2 = 0.1R \Rightarrow R = 20\%$.

[New Practice]

Q14. [NAT · 2 marks]

Find the compound interest on Rs. 8000 at 5% per annum for 2 years, compounded annually.

Answer: 820

Solution: $A = 8000 \times (1.05)^2 = 8820$. $CI = 8820 - 8000 = 820$.

[New Practice]

Q15. [NAT · 1 mark]

Find the amount on Rs. 25000 at 8% per annum for 2 years, compounded annually.

Answer: 29160

Solution: $25000 \times (1.08)^2 = 25000 \times 1.1664 = 29160$.

[New Practice]

Q16. [MSQ · 2 marks]

Which of the following are TRUE about simple interest (SI) and compound interest (CI)?

- (A) For the same P, R, and $T=1$ year, SI and CI are equal
- (B) CI compounds on top of previously accumulated interest; SI does not
- (C) SI grows linearly with time; CI grows exponentially (geometrically) with time
- (D) SI is always greater than CI, for any $T > 1$

Answer: (A), (B), (C)

Solution: (D) is false -- for $T > 1$, CI is generally GREATER than SI, the opposite of what is stated.

[New Practice]

Q17. [NAT · 2 marks]

Find the difference between compound interest and simple interest on Rs. 10000 at 10% per annum for 2 years.

Answer: 100

Solution: $SI = 10000 \times 10 \times 2 / 100 = 2000$. $CI = 10000 \times (1.1)^2 - 10000 = 2100$. Difference = $2100 - 2000 = 100$.

[New Practice]

Q18. [MCQ · 2 marks]

Which is a well-known shortcut formula for the DIFFERENCE between CI and SI over 2 years, given principal P and rate R%?

- (A) Difference = $P \times (R/100)^2$
- (B) Difference = $P \times R^2/100$
- (C) Difference = $P \times R/100$
- (D) Difference = $2PR/100$

Answer: (A)

Solution: For $T=2$ years, Difference = $P(R/100)^2$. Check with $P=5000$, $R=10\%$: $5000 \times (0.1)^2 = 5000 \times 0.01 = 50$, matching an earlier computed difference.

[New Practice]

Q19. [NAT · 1 mark]

Using the shortcut formula $\text{Difference} = P(R/100)^2$, verify the difference for $P=10000$, $R=10\%$.

Answer: 100

Solution: $10000 \times (0.1)^2 = 10000 \times 0.01 = 100$.

[New Practice]

Q20. [NAT · 2 marks]

A sum of Rs. 6000 becomes Rs. 7260 in 2 years at compound interest, compounded annually. Find the rate of interest per annum.

Answer: 10

Solution: $7260 = 6000(1+R/100)^2 \Rightarrow (1+R/100)^2 = 7260/6000 = 1.21 \Rightarrow 1+R/100 = 1.1 \Rightarrow R=10\%$.
